

Bo Liu

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Experience: 12 Years

Target Roles: AI Agent Engineer / AI Application Engineer / LLM Engineer

Expected Salary: Negotiable

Education: Bachelor's Degree

Professional Summary

Experienced engineer specializing in AI Agent systems and LLM-based applications, with end-to-end delivery capability from requirement analysis to production deployment. Proven expertise in RAG pipelines, knowledge graphs, multi-agent orchestration, and LLM optimization. Strong background in building scalable, production-grade systems with measurable performance improvements.

Core Skills

- **AI Stack:** FastAPI, RAG, Agent, LangGraph, Knowledge Graph (Neo4j)
 - **Retrieval Systems:** Elasticsearch, Milvus, Hybrid Retrieval (RRF + Rerank)
 - **LLM Engineering:** Qwen2.5, LoRA Fine-tuning, Function Calling, Tool Integration
 - **Data Processing:** PDF Parsing, OCR, Structured Extraction, Embedding
 - **Infrastructure:** Docker Compose, vLLM, GPU Deployment
 - **Testing & QA:** Automation Frameworks, Performance Testing, Fault Analysis
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GitHub

- <https://github.com/lobokiol/InitialArchitectureForMedicalAgent1>
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Work Experience

Chinasoft International Technology Co., Ltd.

AI Agent Engineer

Jul 2024 – Present

Project 1: Intelligent Medical Triage Agent System

Overview

Developed a hospital triage system based on RAG, Agent architecture, Knowledge Graph, and hybrid retrieval. The system supports multi-turn diagnosis, symptom extraction, and department recommendation.

Key Responsibilities

- Designed a multi-turn dialogue state machine using LangGraph, supporting intent recognition, slot filling, dynamic questioning, and answer validation

- Implemented knowledge graph reasoning pipeline: symptom → disease → department using Jaccard similarity and score aggregation
- Built hybrid retrieval system combining Elasticsearch and Milvus, with RRF fusion and reranking
- Developed LLM-based modules for symptom extraction, semantic normalization, and response generation using Qwen2.5
- Designed risk control mechanisms including negation detection and emergency rules (100+ rules)
- Implemented confidence evaluation combining KG and RAG scores with threshold-based decision logic
- Built PDF ingestion pipeline: parsing, OCR, structured extraction, chunking, and validation
- Deployed system locally using Docker Compose, vLLM, and GPU-based inference
- Designed modular Skills layer including symptom processing, KG reasoning, RAG retrieval, and follow-up generation
- Led multi-modal extension (image + text) using structured engineering workflow (SPEC → PLAN → BUILD → TEST → REVIEW → SHIP)

Tech Stack

LangGraph, Neo4j, Elasticsearch, Milvus, Qwen2.5 7B/72B, bge-large-zh, vLLM, FastAPI

Achievements

- Covered ~80% of common triage scenarios, reducing manual workload
 - Improved retrieval accuracy by 15%–20%, with Top-5 recall ~75%
 - Controlled average dialogue turns to 3–5
 - Completed 10,000+ automated test interactions, with <5% system error rate
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Project 2: Medical Intent Classification (LoRA Fine-tuning)

Overview

Developed a high-accuracy intent classification module for triage entry, supporting structured JSON output.

Key Responsibilities

- Designed classification schema: symptom / process / mixed / non-medical / emergency
- Fine-tuned Qwen2.5-7B using LoRA for structured extraction
- Built training dataset with 2,000 labeled samples including hard cases
- Configured training pipeline on L40S GPU

Training Configuration

- LoRA: r=16, alpha=32
- Target modules: full projection layers
- Learning rate: 1e-4
- Batch size: 16
- Epochs: 3

Achievements

- Reduced GPU memory usage by >70% using LoRA
 - Achieved >95% classification accuracy
 - Improved robustness on negation and implicit expressions by ~30%
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Chinasoft International Technology Co., Ltd.

Software Test Engineer (Cloud Storage SaaS)

Oct 2022 – Jun 2024

- Rebuilt core automation framework, improving stability by ~60%
 - Designed failure analysis and one-click retry system
 - Led P0/P1 defect resolution and quality assurance processes
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Kr Network Technology Co., Ltd.

Software Test Engineer

Jun 2017 – Jun 2022

- Developed automated testing systems using Python, Selenium, Pytest
 - Conducted performance testing using JMeter and Locust
 - Covered functional and API testing
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Shanghai Dasheng Mould Co., Ltd.

Electrical Engineer

Jul 2014 – Jun 2017

- Developed PLC programs and HMI interfaces for industrial automation
 - Designed communication protocols between control systems
 - Performed system debugging, fault diagnosis, and optimization
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Education

Yuncheng University

Bachelor's Degree in Electronic Science and Technology

2010 – 2014

Additional Notes

- Strong experience in production-grade AI systems
- Capable of independent system architecture and delivery
- Focus on reliability, scalability, and cost optimization